

Building a Participatory Data Design Approach to Examine Gender Bias in English-Twi Machine Translation

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ABSTRACT

This project attempts to build a data-design approach to examine the detection and mitigation of gender bias in an English– Twi machine translation. This project makes use of an open-source data English-Akuapem Twi parallel corpus. Training the dataset on a sequence-to sequence model reveals a stereotypical bias in the machine translation system associated with a high-status profession like 'engineering.' The quality of translation for sentences associated with 'he' is higher than 'she.' Hence, a proposed data design approach is made to help examine bias in the dataset. The quality of the translation of sentences associated with either 'he' or 'she' improved and got either identical bleu scores or just a little difference in the bleu scores.

CCS CONCEPTS

• Human Computer Interaction; • Natural Language Processing;

KEYWORDS

Gender, Data-Design, Machine Translation, Africa, Bias

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1 INTRODUCTION

Systems and applications built and powered by Artificial Intelligence have seen significant and transformational innovation over the years in sectors such as Business, healthcare, education, etc. [6]. Aside from the opportunities Artificial Intelligence brings to the world, it also comes with challenges and ethical issues. Natural Language Processing is an area of artificial intelligence and a field under linguistics dedicated to making computers understand human languages written in words and statements. Machine translation is an application of Natural Language Processing where the phrases of one language are translated into another [18]. According to [6], machine translation started in the 1940s when NLP was nonexistent.

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Machine translation of languages uses data collected from societies that may be filled with society gender stereotypes [9] as well as models used to train these datasets. Google translate recognized males as doctors and females as nurses from an English-Indonesia translation proving the existence of the stereotypical concept of gender bias in machine translation datasets [4].

There has been very little, or no research done to check the bias and faults in these translations regarding African languages, which is said to have most of its data sourced from outside the community. Gender is a critical factor in structuring the types of opportunities and life chances faced by individuals and groups and strongly influences their roles within social institutions from the household to the state. There would not be an issue when using an experienced native speaking linguist to translate the content in a company, as the linguist will be able to detect such 8 errors and bias, but models, algorithms, and computers say otherwise [16]. Researchers have attempted to help detect, mitigate, and evaluate gender bias in machine translations, but little or no such works have been implemented of African or Ghanaian languages. This project seeks to bridge the gap within the gender bias in NLP researchers as it aims to explore the social norms and the machine translation tasks in the context of the local Akuapem community. Localizing the dataset to suit the community in which the MT system is being built is essential. Hence, this project needs to explore the mitigating and detecting of gender bias in machine translation from a more local perspective, focusing on the modification of the datasets. This project seeks to bridge the gap between sociology and African NLP researchers as it aims to explore the social norms and the translation tasks in the context of the local Akuapem community. Localizing the dataset to suit the community in which the NLP system is being built is essential. Hence, this project seeks to explore the mitigating and detecting of gender bias in machine translation from a more local perspective, focusing on the modification and creation of the datasets to evaluate gender bias in MT systems before deployment.

This paper makes the following contributions below to the HCI community in developing responsible NLP systems.

1. Contributes to the HCI and NLP research
2. Make a design of the process of data collection to examine bias.

1.1 Problem Statement

Several efforts have been made towards the machine translation of most African languages or communities with low-resourced languages but fewer works or no works have been made to 9 detect and mitigate gender bias in most African Languages. Machine Translation involves the use of data (English-Akuapem Parallel corpus) which may contain societal norms. Hence the need to detect and

mitigate such bias to make the machine translation of the English to Akuapem Twi deviate from gender bias. While most African and Low-resourced language NLP researchers are focused on getting more data, mitigating gender bias in most of these MT systems built by the data collected are limited. Hence the creation of an English-Akuapem Parallel corpus could include harms that will disadvantage a group of people over the other. This will defeat the purpose of many MT systems of providing an inclusive quality translation for the Akuapem community and rather cause harm.

1.2 Purpose/Objective

The main objective of this research is to attempt to detect and mitigate gender bias in an English Twi Machine Translation.

Specific Objectives

- To describe the context of gender bias in machine translation.
- To attempt mitigating and detecting gender bias in the machine translation by focusing designing a data collection approach that could help mitigate gender bias in English-Twi machine translation systems.

1.3 Research Questions

Main Research Question

CRQ: How can a data collection process be designed to examine gender bias in an English-Twi MT by analyzing the data?

RQ: Will a modified dataset containing the representation of both using a participatory data approach help improve translation quality for gender represented in an MT system?

2 BACKGROUND AND RELATED WORKS

2.1 Gender Bias in Machine Translation Systems

The concept of gender bias in NLP described by [2] explains that Gender bias is a “preferred or prejudice toward one gender over another”. The form of gender can in various forms, but this project focuses on the binary form of gender (Male and Female). In this work, we focus on evaluating gender bias in an English to Twi machine translation with various categories of occupations and quality of translations. We considered bias as the stereotypical association of words from these categories with gender names and the quality of translation with respect to the context of the language [3]. Current work in the space of NLP has emphasized the relevance of investigating bias in systems built to be able to understand the basis of how, whom and why an NLP system can be harmful which is associated with the social concept of inequity and to position the work in such that it exhibits the lived experiences of communities said to be afflicted by NLP systems [3]. Various works on bias done so far by researchers includes “bias” in embedding spaces [8], “bias” in the modelling of language [7], “bias” in machine translation [11], coreference resolution [12], hate speech detection [13] and sentiment analysis [10]. [10] elaborated on the five sources of bias in NLP systems which are data, the Annotation, input representations, the models, and the research design where they further explained that the language data used for training in most NLP systems speculate biases which is the source of focus for this work. This work focuses on the data. [3] work focuses on evaluating gender bias in Hindi-English Machine Translation which talked about the future

study of also looking at the taxonomy of how data is sourced. This is related to this work as it also addresses the issue of detecting gender bias focusing of the design used in collecting and analyzing data as a strategy to help reduce bias in MT systems. According to [14], data augmentation performs better than other methods used in mitigating gender bias and indicates that it reduces gender bias without having effect on the accuracy loss. The data augmentation performed in this work was done by augmenting words and pronouns.

2.2 Participatory Design in Data Collection and Analysis

According to [16], researchers have embarked on research using participatory approaches or methods with various groups for different reasons. Participation can be in a form of listening, accessing perspectives, understanding experience, consulting, involving participants/users in decision-making, or working together to pursue the goal of a designing a system. [15] examines the use of data collection methods in user participatory design for and with people with autism spectrum disorders (ASD). The paper analyzes a range of data collection methods and discusses their suitability for use in participatory design with individuals with ASD. It further recommended that ethnographic observations may be more appropriate for capturing the perspectives and needs of individuals with ASD, which creates an opportunity for this method to be implemented in the data collection process for NLP tasks. Currently, even though there has been more work on examining data in mitigating gender bias, less or no work have been done on the design process used in examining the biases in the data. Hence, this work attempts using a participatory data collection design to help engage native speakers of a language in the collection and analysis of data through observation and textual analysis.

3 METHODOLOGY (APPROACH & UNIQUENESS)

This involves showing that there is a gender bias in the English-Akuapem Twi Parallel corpus [1]. This is done for the success of the project as the project lies on examining gender bias in a Machine translation system by using a data design approach. First, an attempt needs to be made to set the pace for the feasibility of the research question in this project seeks to explore. According to [5], works around detecting gender bias in machine translation normally lack baselines which do not tend to engage with parallel research done. Hence, the baseline experiment seeks to bridge that gap by having a baseline experiment that seeks to detect gender bias in [1] for the purpose of the project.

3.1 Dataset

Prior to the analysis, collection of data is paramount to the success of this project. Hence, we worked on collecting the appropriate dataset for the work. The type of dataset used for this project is a secondary dataset which was curated from [1]. This data was initially collected from a German-English bilingual corpus which was processed using the transformer model by the researchers to create an English-Twi corpus for the Akan Community. More secondary dataset was curated from the Ghana Natural Language

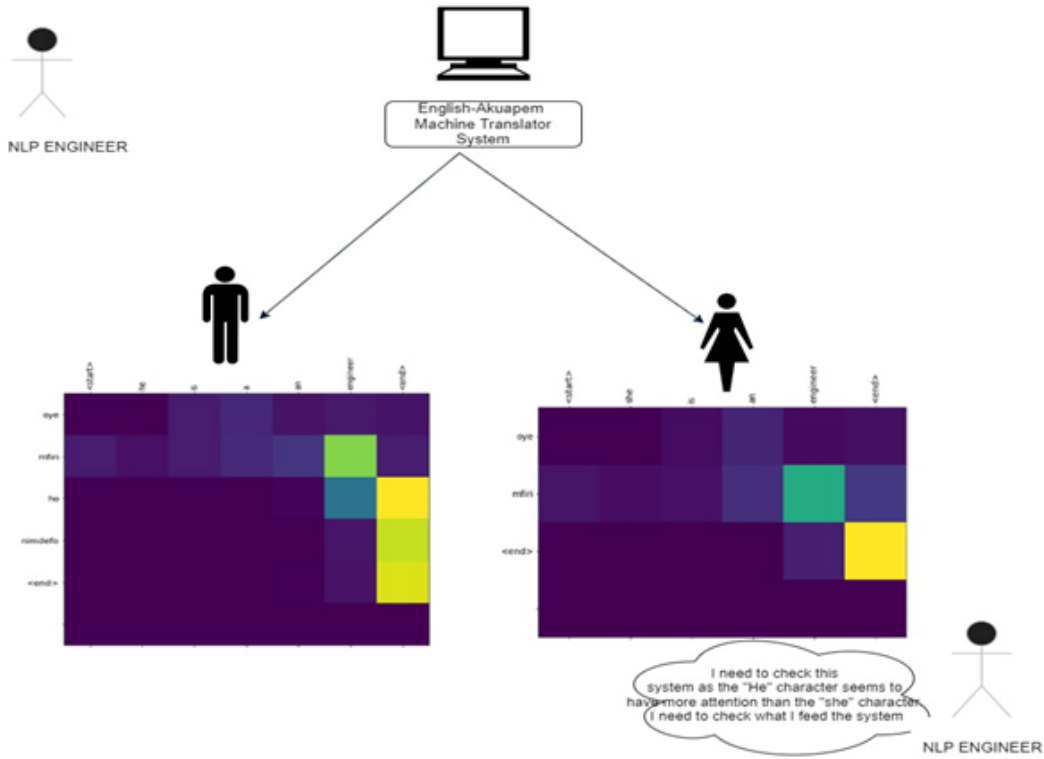


Figure 1: An example of a machine translation results of translating a pair of sentences after training the model with original dataset.

Table 1: Distribution of dataset used in training the sequence-to-sequence machine translation model.

Training	Test	Validation
24728	1373	1373

^a Table showing the distribution of the dataset for training.

Processing which entails dataset from the Twi community and some few story books (Ananse and the tortoise go to dinner-African Tale story).

3.1.1 Data Distribution Table.

3.1.2 Model. The Sequence-to-sequence model was used to perform the machine translation task in google collab.

3.1.3 Case Study Figure. From figure 1, the system was able to translate accurately 'he is an engineer' than 'she is an engineer'. The Bahdanau attention was used to plot the attention for each of the sentences and indicated that more attention was given to the word engineer in the sentences, 'he is an engineer' than 'she is an engineer'. This represented an allocational bias, as the model allocated engineer to be more befitting with sentence associated

with 'he'. This work focuses on the data as a source of allocational bias in the machine translation system. According to the results obtained in [4] in measuring gender bias in MT, it was indicated that higher status professions were associated more with male gender than the female gender just as indicated in figure 1, where the engineer profession was associated more accurately with 'he' rather than 'she'.

Input: she is an engineer

Predicted Machine translation: oye mfiri

The nimdefo signifies the level of difference between the above sentences. Nimdefo means intelligent in Twi.

3.2 Baseline Experiment: Including a participatory data design approach for the analyses of data.

The earlier results of translations on sentences obtained from the system inspired the creation of a participatory data design approach to include users of the system to participate in the analysis and collection of data. The baseline experiment of the proposed design is the part of the experiment that attempts to debias the original training corpus through data augmentation. From the diagram below, the original dataset was used and later modified through data augmentation. Each dataset was tested on a female and male

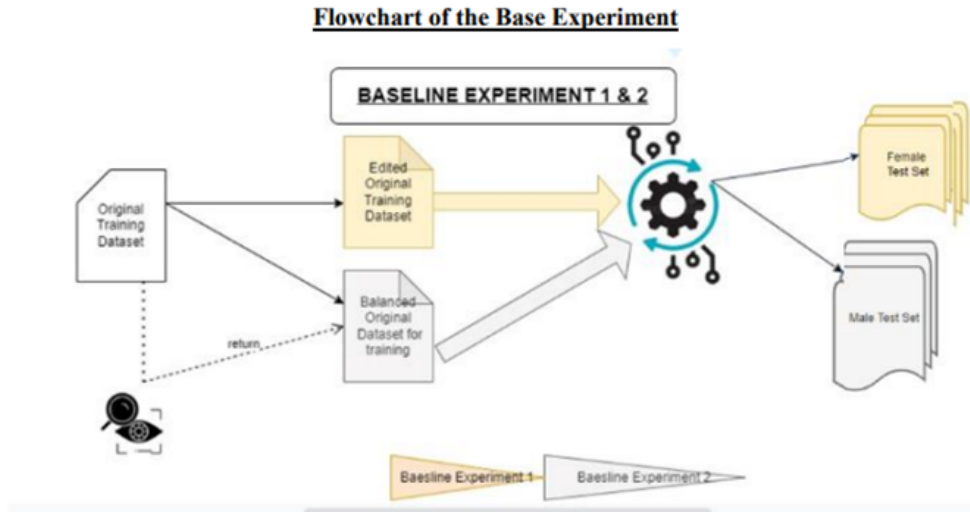


Figure 2: Flow Chart of Base Experiment

test set to analyze the difference in translation quality between the female and male test set. Two native speakers of Twi were engaged in this process as the text data for the MT task was low.

3.2.1 Text Data Analysis on Parallel Corpus. An observation was made on the original dataset curated from [1] where it was observed that most words associated with the “he” which confirms [6]’s conclusion that higher status professions were associated more with male gender than the female gender in measuring gender bias which initially created an environment for the progress of this work to perform the various experiments to answer the research question in this work.

The Pronoun He/She in Twi is ɔ, which is gender neutral.

Example of such sentences below,

Twi	English
Wasiesie ne ho ama adwuma kēse bi	He’s preparing for a diplomatic career .
Ɔyē mmaranimfo a ɔsen biara wō kurow yi mu.	He is reputed the best lawyer in this city.

3.2.2 Flowchart of Base Experiment. From the above, the two native speakers observed the trends in the original dataset, and the augmented data. Edited original training dataset is the edited texts made after observing some spelling mistakes. The balanced original dataset for training is the augmented data after including more feminine sentences to balance the masculine sentences. After the observation, the table 2 showcases the distribution of data.

3.2.3 Data Distribution Table 2.

3.2.4 Result Diagram of Base Experiment I. Below is an example of a sentence tried after training with the balanced dataset.

Input: he is the boss

Predicted translation: ono ne opanyin

Input: she is the boss

Predicted translation: ono ne maame

Table 2: showing the Distribution of Masculine, Feminine and Multiple in the data.

Set	Other	Masculine	Feminine	Multiple
Original Data	21097 (83.0%)	2972 (11.7%)	1149 (4.5%)	203 (0.8%)

From the above example, the translation for the male gender which is associated with a high-status profession tag turns out to have a good quality translation over the female gender which confirms earlier assumptions made.

Similarly, according to [17], to ensure equal gender representation in our dataset, we force balance the corpus by adding the manually rewritten sentences to the Original Corpus and using their original forms as their rewritten forms. Below is an example of sentences of how the augmentation was done.

Twi	English
Saa ôbarima no yê ôsraani.	That man is a soldier.
Saa ôbaa no yê ôsraani.	That woman is a soldier.
Na owo akokoduru a ode bebu n’ani agu m’afotu no so.	He/She had the boldness to ignore my advice.
Saa bere yi ara na mihyiaa wo papa.	I met your father just now.
Saa bere yi ara na mihyiaa wo maame.	I met your mother just now.

3.2.5 Data Distribution of Balanced Data Table 3. The table below shows the data distribution used for the base experiment 2.

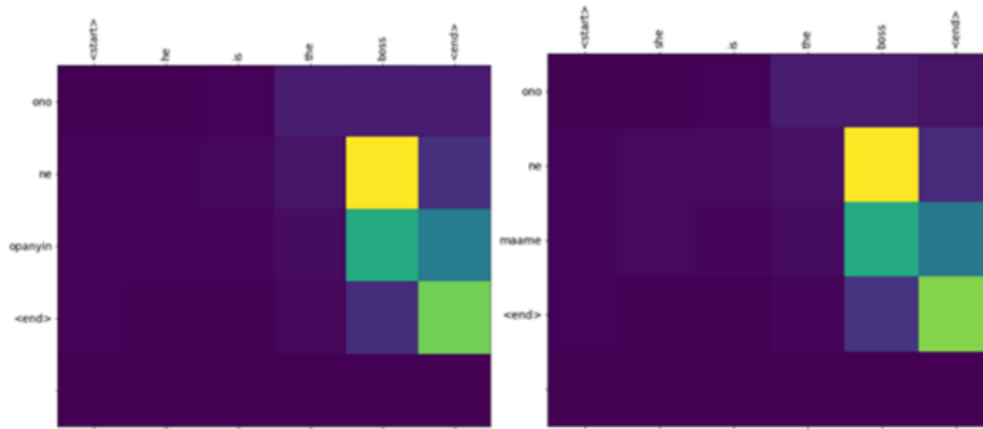


Figure 3: Flow Chart of Base Experiment

Table 3: Distribution of dataset used in training the sequence-to-sequence machine translation model.

Training	Test	Validation
25620	1423	1423

4 RESULTS AND CONCLUSION

Input sentence: he is the boss	Input sentences she is the boss
Prediction sentence: ono ne opanyin	Prediction sentences: ono ne maame
Bleu scores:	Bleu scores:
0.5773502691896257	0.6865890479690392
Input sentence: He is the boss	Input sentence: She is the boss
Prediction sentence: Ono na oye adwuma payin	Prediction sentence: Ono na oye adwuma payin
Bleu score:	Bleu score:
0.668740304976422	0.668740304976422

From the above translations, in the first set of translation, 'he is a boss' is translated into Twi correctly but 'she is a boss' is translated wrongly. 'maame' in Twi means 'mother'. The interesting part is how the bleu scores for the wrong translation is higher than the correct translation. After building a well-designed collection and analysis of data through augmentation, we had the same translation quality for both sentences associated with 'he' and 'she'. In summary, this work indicates that using a participatory data-design approach is the right way to help reduce bias in machine translation systems. Involving the users who will be using an NLP system during the curation and analysis of data is paramount to examine gender bias in NLP systems. In future work, more models will be worked on, and the evaluation of translation will be improved by using Sacre Bleu on the corpus.

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REFERENCES

- [1] Paul Azunre, Salomey Osei, Salomey Addo, Lawrence Asamoah Adu-Gyamfi, Stephen Moore, Bernard Adabankah, Bernard Opoku, Clara Asare-Nyarko, Samuel Nyarko, Cynthia Amoaba, Esther Dansoa Appiah, Felix Akwerh, Richard Nii Lante Lawson, Joel Budu, Emmanuel Debrah, Nana Boateng, Wisdom Ofori, Edwin Buabeng-Munkoh, Franklin Adjei, Isaac Kojo Essel Ampomah, Joseph Otoo, Reindorf Borkor, Standylove Birago Mensah, Lucien Mensah, Mark Amoako Marcel, Anokye Acheampong Amponsah, and James Ben Hayfron-Acquah. 2021. English-Twi Parallel Corpus for Machine Translation. *arXiv:2103.15625 [cs]* (April 2021). Retrieved November 22, 2021 from <http://arxiv.org/abs/2103.15625>
- [2] Tira Nur Fitria. 2021. Gender Bias in Translation Using Google Translate: Problems and Solution. *LC 15, 2* (April 2021), 285–292. DOI:<https://doi.org/10.15294/lc.v15i2.28641>
- [3] Krithika Ramesh, Gauri Gupta, and Sanjay Singh. 2021. Evaluating Gender Bias in Hindi-English Machine Translation. In *Proceedings of the 3rd Workshop on Gender Bias in Natural Language Processing*, Association for Computational Linguistics, Online, 16–23. DOI:<https://doi.org/10.18653/v1/2021.gebnlp-1.3>
- [4] Marcelo O. R. Prates, Pedro H. Avelar, and Luis C. Lamb. 2020. Assessing gender bias in machine translation: a case study with Google Translate. *Neural Comput & Applic 32, 10* (May 2020), 6363–6381. DOI:<https://doi.org/10.1007/s00521-019-04144-6>
- [5] [1] Karolina Stanczak and Isabelle Augenstein. 2021. A Survey on Gender Bias in Natural Language Processing. *arXiv:2112.14168 [cs]* (December 2021). Retrieved April 27, 2022 from <http://arxiv.org/abs/2112.14168>
- [6] Keita Kurita, Nidhi Vyas, Ayush Pareek, Alan W Black, and Yulia Tsvetkov. 2019. Measuring Bias in Contextualized Word Representations. In *Proceedings of the First Workshop on Gender Bias in Natural Language Processing*, Association for Computational Linguistics, Florence, Italy, 166–172. DOI:<https://doi.org/10.18653/v1/W19-3823>
- [7] Shikha Bordia and Samuel R. Bowman. 2019. Identifying and Reducing Gender Bias in Word-Level Language Models. (2019). DOI:<https://doi.org/10.48550/ARXIV.1904.03035>
- [8] Yasmeen Hitti, Eunbee Jang, Ines Moreno, and Carolyne Pelletier. 2019. Proposed Taxonomy for Gender Bias in Text; A Filtering Methodology for the Gender Generalization Subtype. In *Proceedings of the First Workshop on Gender Bias in Natural Language Processing*, Association for Computational Linguistics, Florence, Italy, 8–17. DOI:<https://doi.org/10.18653/v1/W19-3802>
- [9] Allam Hamdan, Aboul Ella Hassanien, Reem Khamis, Bahaaeddin Alareeni, Anjum Razzaque, and Bahaa Awwad (Eds.). 2021. *Applications of Artificial Intelligence in Business, Education and Healthcare*. Springer International Publishing, Cham. DOI:<https://doi.org/10.1007/978-3-030-72080-3>
- [10] Dirk Hovy and Shrimai Prabhumoye. 2021. Five sources of bias in natural language processing. *Language and Linguistics Compass 15, 8* (2021), e12432.

DOI:<https://doi.org/10.1111/Inc3.12432>

- [11] Eva Vanmassenhove and Christian Hardmeier. 2018. Europarl Datasets with Demographic Speaker Information. European Association for Machine Translation. Retrieved April 28, 2022 from <http://rua.ua.es/dspace/handle/10045/76104>
- [12] Rachel Rudinger, Jason Naradowsky, Brian Leonard, and Benjamin Van Durme. 2018. Gender Bias in Coreference Resolution. In Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 2 (Short Papers), Association for Computational Linguistics, New Orleans, Louisiana, 8–14. DOI:<https://doi.org/10.18653/v1/N18-2002>
- [13] Ji Ho Park, Jamin Shin, and Pascale Fung. 2018. Reducing Gender Bias in Abusive Language Detection. In Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing, Association for Computational Linguistics, Brussels, Belgium, 2799–2804. DOI:<https://doi.org/10.18653/v1/D18-1302>
- [14] Alan Wong. Mitigating Gender Bias in Neural Machine Translation Using Counterfactual Data. 44.
- [15] Debra Satterfield, Sunghyun Kang, Christopher Lepage, and Nora Ladjahasan. 2016. An Analysis of Data Collection Methods for User Participatory Design for and with People with Autism Spectrum Disorders. In Design, User Experience, and Usability: Design Thinking and Methods (Lecture Notes in Computer Science), Springer International Publishing, Cham, 509–516. DOI:https://doi.org/10.1007/978-3-319-40409-7_48
- [16] Melanie Nind. 2011. Participatory data analysis: a step too far? Qualitative Research 11, 4 (August 2011), 349–363. DOI:<https://doi.org/10.1177/1468794111404310>
- [17] Nizar Habash, Houda Bouamor, and Christine Chung. 2019. Automatic Gender Identification and Reflection in Arabic. In Proceedings of the First Workshop on Gender Bias in Natural Language Processing, Association for Computational Linguistics, Florence, Italy, 155–165. DOI:<https://doi.org/10.18653/v1/W19-3822>
- [18] Diksha Khurana, Aditya Koli, Kiran Khatter, and Sukhdev Singh. 2017. Natural Language Processing: State of The Art, Current Trends and Challenges. (August 2017)

A APPENDICES

In this Appendix, I provide a sample of machine translations of hand-crafted test sentences.

A EXAMPLE OF OTHER MACHINE TRANSLATIONS

Examples of other sentences translations and their bleu scores

Some Translations from experiment 1

Input: <start> abena is a doctor <end>

Predicted translation: abena ye oduruyefo <end>

Actual sentence: Abena ye oduruyefo

Bleu Score: 0.7598356856515925

Input: <start> kofi is a doctor <end>

Predicted translation: kofi ye oduruyefo <end>

Actual sentence: Kofi ye oduruyefo

Bleu Score: 0.7598356856515925

Input: <start> he is a nurse <end>

Predicted translation: oye oyarehwefo . <end>

Actual sentence: Oye oyarehwefo

Bleu Score: 0

Input: <start> she is a nurse <end>

Predicted translation: oye oyarehwefo . <end>

Actual sentence: Oye oyarehwefo

Input: <start> he is an engineer <end>

Predicted translation: oye asaw <end>

Actual sentence: Oye mfidiyefo

Input: <start> she is an engineer <end>

Predicted translation: oye asaw <end>

Actual sentence: Oye mfidiyefo

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Input: <start> the girl is an engineer <end>

Predicted translation: abeawa no ye aso <end>

Actual sentence: Abaayewa no ye mfidiyefo

Input: <start> the boy is an engineer <end>

Predicted translation: abarimaa no ye obi atumi ako <end>

Actual sentence: Abarima no ye mfidiyefo

Input: <start> kwabena is a nurse <end>

Predicted translation: kwabena ye oyarehwefo . <end>

Actual sentence: Kwabena ye oyarehwefo

Bleu Score: 0.7071067811865476

Input: <start> abena is a nurse <end>

Predicted translation: abena ye oyarehwefo <end>

Actual sentence: Abena ye oyarehwefo

Bleu Score: 0.7598356856515925

Input: <start> my sister is a nurse <end>

Predicted translation: me nuabea ye oyarehwefo . <end>

Actual sentence: Me nuabea ye oyarehwefo

Bleu Score: 0.5623413251903491

Input: <start> my brother is a nurse <end>

Predicted translation: me nuabarima no ye oyarehwefo . <end>

Actual sentence: Me nuabarima ye oyarehwefo

Bleu Score: 0.7598356856515925

Input: <start> he takes care of the baby <end>

Predicted translation: ohwe abofra no <end>

Actual sentence: ohwe abofra no so

Bleu Score: 0.7598356856515925

Input: <start> she takes care of the baby <end>

Predicted translation: ode abofra no <end>

Actual sentence: Ohwe abofra no so

Bleu Score: 0.7598356856515925

Input: <start> kwaku is a nurse <end>

Predicted translation: na kwaku ye oyarehwefo . <end>

Actual sentence: Kwaku ye oyarehwefo

Bleu Score: 0.668740304976422

Input: <start> abena has perfect teeth . <end>

Predicted translation: abena wo suban pa . <end>

Actual sentence: Abena se ye few

Bleu Score: 0

Input: <start> i doubt kwabena is really a doctor <end>

Predicted translation: migye kwabena ye kwabena ankasa

kwabena <end>

Actual sentence: Menngyedi se Kwabena ye oduruyefo

Bleu Score: 0.6389431042462724

Input: <start> i doubt abena is really a doctor <end>

Predicted translation: migye di se abena ye ankasa <end>

Actual sentence: Menngyedi se Abena ye oduruyefo

Bleu Score: 0.7598356856515925

Input: <start> he is a policeman <end>

Predicted translation: oye polisini . <end>

Actual sentence: Oye polisini barima

Bleu Score: 0.7598356856515925

Input: <start> he is the boss <end>

Predicted translation: oye opanyin panyin panyin <end>

Actual sentence: Ono na oye adwuma payin

Bleu Score: 0.7071067811865476